PROJECT DOCUMENTATION | POWER BI

T-Shirt Pricing & Catalog Intelligence Dashboard

Catalog health · Pricing · Brand mix

Prepared by **Tahseenah Hamza**, Data Analyst and Power BI Developer

tahseenahhhamza@gmail.com | github.com/Tahseenahhamza

Products (SKUs)	Brands	Report pages	DAX measures
1,428	143	3	12

1. Title and Introduction

Project name: T-Shirt Pricing & Catalog Intelligence Dashboard (Power BI, June 2026)

The business problem: Online shops show big discounts. But a discount is only real if the starting price (called MRP) is fair, and the shop still makes money at the selling price. This project studies men's t-shirt prices. It shows where pricing is healthy and where it depends on high MRPs and big discounts.

My role: Data Analyst and Power BI Developer. I built the data model, the formulas (DAX), and the 3 report pages. I also checked the numbers and wrote this document.

Objectives

- Find out how far the MRP is above the real selling price.
- Compare profit and discount across 5 price tiers (Luxury, Premium, Mid-range, Economy, Budget).
- Show which brands sell the most products and if they also earn the most.
- Give buying and pricing teams a simple, filterable view to make better decisions.

2. Problem Statement

The main challenge: The average discount is above 50%. So a big discount tells us very little. Teams need to know which prices are fair, which MRPs are too high, and which tiers and brands really earn money.

Questions this project answers

- How big is the gap between MRP and the selling price?
- Which price tiers earn good profit per unit? Which ones depend on big discounts?
- How many products have a discount of 70% or more?

- Which brands sell the most products? Do they earn the most too?
- Do international brands do better or worse than domestic brands?

Who needs this, and why

Who	What they need
Category / buying team	Which tiers and brands to grow, cut, or negotiate.
Pricing team	Where MRP and discounts look too high or not sustainable.
Marketing team	Which offers are believable and which are not.
Leadership	A quick view of product mix and profit health.

Why it matters: 26.5% of products have a discount of 70% or more. If those MRPs are not believable, the catalog is weaker than the discount banners suggest.

3. Data Methodology

Source: Product listing data for men's t-shirts from an Indian e-commerce platform. It has 1,428 products from 143 brands. I also used a Brand Tier table that shows the tier and origin of each brand.

Data model (star schema)

Table	Type	Rows	Main columns
T-Shirt Products	Fact	1,428	Brand, Title, Sale Price, MRP, Cost Price
Brand Tier Reference	Dimension	143	Brand, Tier, Origin, Type (Domestic / International)

The two tables are joined using the **Brand** column. So every product gets its brand's tier and type.

Calculated columns (6)

Column	How it is worked out
Profit Per Unit	Sale Price minus Cost Price
Profit Margin %	Profit Per Unit divided by Sale Price
Discount Amount	MRP minus Sale Price
Discount %	Discount Amount divided by MRP
Price Band	Puts each product in a sale-price range (used for filters)
Discount Flag	Shows "High Discount" when Discount % is 70% or more

Steps and accuracy checks

- I loaded both tables into Power BI and built a star schema.
- I wrote 12 DAX measures in 4 folders: Core Metrics, Profitability, Discount Analysis, and Brand Intelligence.
- I checked every number by running DAX queries and comparing them with the visuals.
- This check found one mistake. The Domestic vs International chart showed MRP instead of Sale Price. I fixed it.

4. Key Performance Indicators (KPIs)

Total Products	Total Brands	Avg Sale Price	Avg Discount %
1,428	**143**	**₹1,034**	**57.1%**
MRP Inflation Ratio	Est. Avg Profit Margin	Avg Profit Per Unit	Extreme Discount (70%+)
2.32x	**26.3%**	**₹243**	**26.5%**

Why I chose these KPIs

Group	KPIs	Why
Catalog size	Total Products, Total Brands	Show how big and how wide the catalog is.
Pricing	Avg Sale Price, MRP Inflation Ratio	Show what customers really pay and how far MRP is above it (average MRP divided by average sale price).
Discounts	Avg Discount %, Extreme Discount %	Show the normal discount and how many products depend on 70%+ discounts.
Profit	Est. Profit Margin, Profit Per Unit	Show if the selling price still earns money. Margin is an estimate because it uses the Cost Price column.

5. Data Analysis and Visualization

Finding 1: Discounts are big, and MRP is far above the selling price

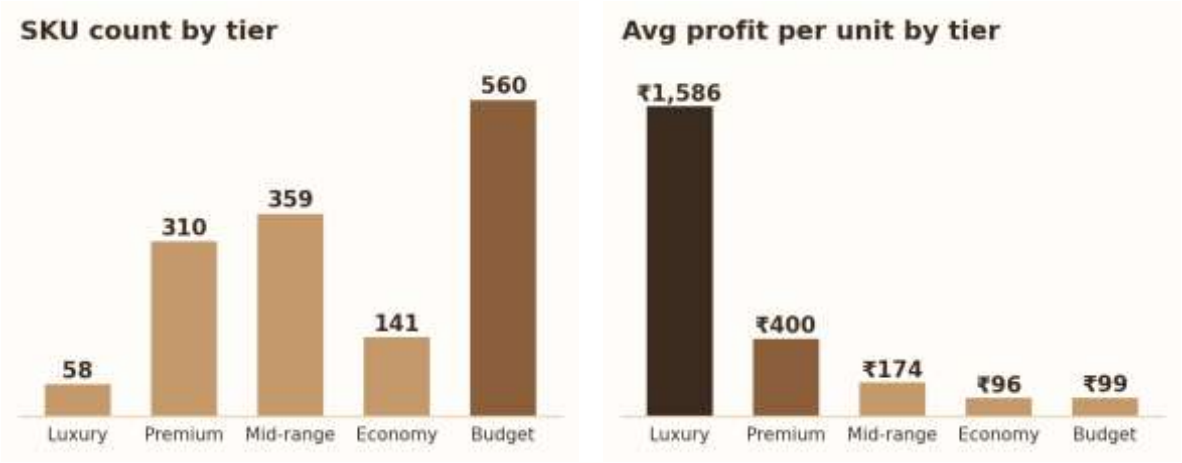


The average discount is **57.1%**. The average MRP is **2.32 times** the average sale price. About **1 in 4 products (26.5%)** has a discount of 70% or more.

Example: the brand iVOC lists a t-shirt at ₹3,499 and sells it at ₹385. That is an 89% discount.

What this means: The discount % mostly shows how high the MRP was set. Sale price and profit are better to look at.

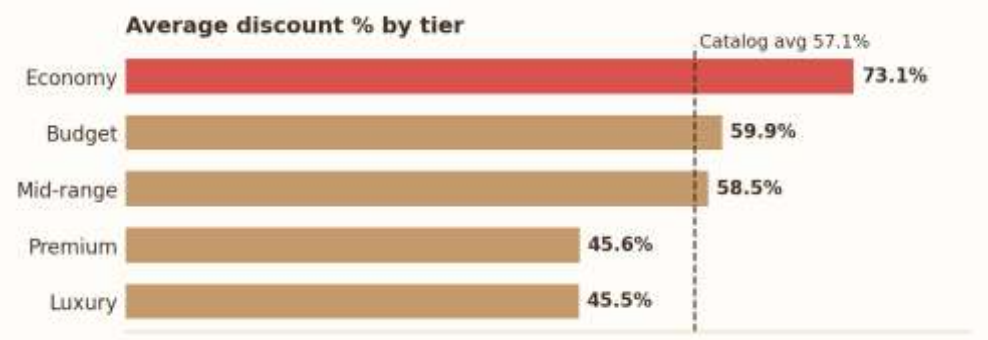
Finding 2: Most products are in the tiers that earn the least



WHAT THE DATA TELLS US

- **Budget** has 560 products (39% of the catalog) but earns only **₹99** per unit.
- **Luxury** has only 58 products (4%) but earns **₹1,586** per unit. That is about 16 times more than Budget.
- Premium has 310 products and earns ₹400 per unit. Mid-range has 359 products and earns ₹174.
- Economy and Budget together have 701 products (almost half) and each earns less than ₹100 per unit.

Finding 3: The Economy tier depends most on big discounts



WHAT THE DATA TELLS US

- Economy has an average discount of **73.1%** and earns only ₹96 per unit.
- Premium and Luxury give about 45.5% discount and earn much more per unit.
- Budget (59.9%) is not the most discounted tier. The big discounts are in the Economy tier.

Finding 4: International brands earn about 4 times more per unit



WHAT THE DATA TELLS US

- Domestic brands have 1,192 products (83.5%) and earn **₹141** per unit.
- International brands have only 236 products (16.5%) but earn **₹575** per unit.
- A small number of international products adds a lot of profit per unit.

Finding 5: The biggest brand is not the most profitable

WHAT THE DATA TELLS US

- **The Indian Garage Co** is the biggest brand with 184 products (12.9% of the catalog) but earns only **₹77** per unit.
- Brand prices are very different. ARMANI EXCHANGE has the highest average price (**₹6,086**). The lowest brand average is **₹285**. That is about 21 times higher.
- Selling the most products does not mean earning the most.

6. Interactivity and Features

The dashboard has 3 pages. Use the buttons on the left side to move between them.

Page	Question it answers	What is on it
Catalog Overview	How healthy is the catalog? How are prices spread?	5 KPI cards, brand price chart, top 15 brands by product count, tier charts
Discount & Profitability	How big are discounts? Do products still earn money?	Margin, discount and profit KPIs, discount by brand and tier, brand table with High Discount flag
Brand Intelligence	How do brands compare on price, profit and discount?	Brand count, top brand, lowest and highest average price, brand table, domestic vs international chart

How to use it

- **Slicers (filters):** Tier, Price Band and Type on the Overview page. Brand, Tier and Price Band on the Brand Intelligence page.
- **Click to filter:** Click a bar, dot or table row and the other charts change to match.
- **Bubble charts:** Each dot is one brand. Size shows the product count. Colour shows the tier.

- **Brand table:** Colours show big discounts (red) and profit (brown). A flag marks 70%+ discounts.
- **Hover:** Move the mouse over any chart to see the exact value.

Example questions you can answer

Question	How to find the answer
How do Premium brands compare on price and discount?	Go to Brand Intelligence and set Tier to Premium.
Which international brands are at the top end?	Go to Overview and set Type to International.
Which brands depend most on big discounts?	Go to Discount & Profitability and sort the brand table by Discount %.
Is a low price band really profitable?	Set Price Band on the Overview page and check profit per unit.

7. Conclusion and Recommendations

Main takeaway: This catalog is built on big discounts, but it earns little on most products. Most products are in the Economy and Budget tiers, which earn less than ₹100 per unit. A small number of Premium, Luxury and international products earn most of the profit per unit.

What the business should do

#	Action	Why (from the data)
1	Judge brands by sale price and profit per unit, not by discount %.	The average discount is 57.1% and MRP is 2.32 times the sale price. So discount % mostly shows the MRP.
2	Move the mix towards Premium and international brands.	They earn ₹400 and ₹575 per unit. Budget earns ₹99 and domestic earns ₹141.
3	Review the Economy tier: change prices, talk to suppliers, or remove weak products.	73.1% average discount and only ₹96 profit per unit.
4	Cut low-earning products before adding new ones.	560 Budget products earn ₹99 each. The biggest brand (184 products) earns ₹77.
5	Set a check for MRP when a product has a discount of 70% or more.	26.5% of products are marked High Discount.

Limits to keep in mind

- This is one snapshot (June 2026). It does not show change over time.
- Profit numbers are estimates. They depend on the Cost Price column in the data.
- A big gap between MRP and sale price suggests the MRP is too high. It does not prove the MRP was never charged.
- Tier comes from the brand table. So one product can be above or below its brand's tier.

- The data is from one platform and has no sales volume.

Next steps: Add sales volume to see how many units sell. Track the same products over time to see how MRP and discounts change.